

LAB-8

Task Description#1

Use AI to generate test cases for `is_valid_email(email)` and then implement the validator function.

Requirements:

- Must contain @ and . characters.
- Must not start or end with special characters.
- Should not allow multiple @.

Task-0

```
def add(x, y):  
    return x + y  
  
def subtract(x, y):  
    return x - y  
  
def multiply(x, y):  
    return x * y  
  
def divide(x, y):  
    if y == 0:  
        return "Error: Division by zero"  
    return x / y  
  
def main():  
    print("Simple Calculator")  
    print("Select operation:")  
    print("1. Add")  
    print("2. Subtract")  
    print("3. Multiply")  
    print("4. Divide")  
  
    choice = input("Enter choice (1/2/3/4): ")  
  
    if choice in ('1', '2', '3', '4'):  
        num1 = float(input("Enter first number: "))  
        num2 = float(input("Enter second number: "))  
  
        if choice == '1':
```

```
def main():
    print("Simple Calculator")
    print("Select operation:")
    print("1. Add")
    print("2. Subtract")
    print("3. Multiply")
    print("4. Divide")

    choice = input("Enter choice (1/2/3/4): ")

    if choice in ('1', '2', '3', '4'):
        num1 = float(input("Enter first number: "))
        num2 = float(input("Enter second number: "))

        if choice == '1':
            print("Result:", add(num1, num2))
        elif choice == '2':
            print("Result:", subtract(num1, num2))
        elif choice == '3':
            print("Result:", multiply(num1, num2))
        elif choice == '4':
            print("Result:", divide(num1, num2))
    else:
        print("Invalid input")

if __name__ == "__main__":
    main()
```

OUTPUT:

Simple Calculator

Select operation:

1. Add
2. Subtract
3. Multiply
4. Divide

Enter choice (1/2/3/4): 2

Enter first number: 58

Enter second number: 72

Result: -14.0

(.venv) PS C:\Users\Rishitha Reddy\OneDrive\Desktop\AIAC\lab-8> █